Advanced_pandas_sales_dataset

QUESTION 1

Q1 Dataset Exploration

Perform the following exploratory steps:

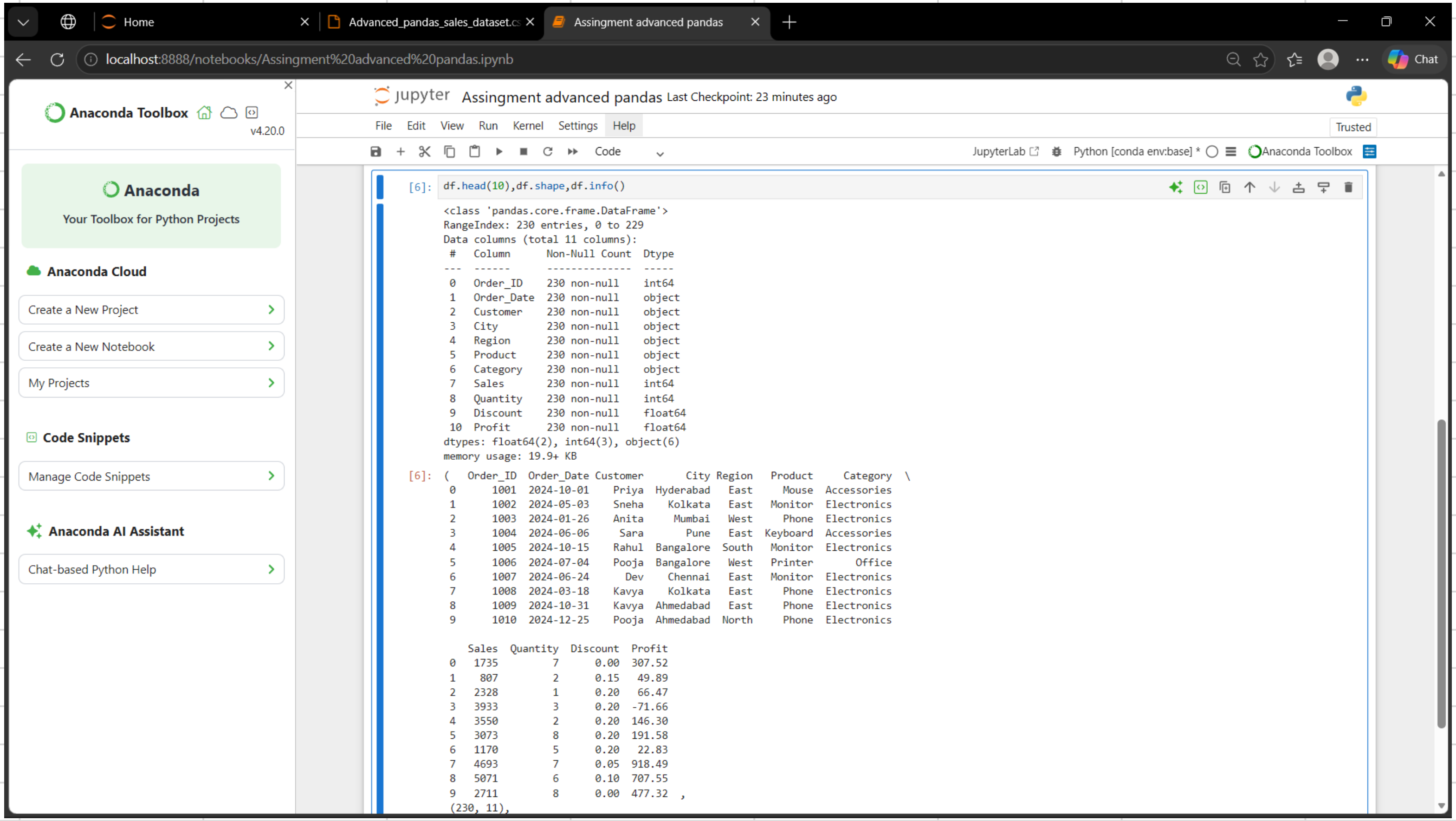
Display the first 10 rows

Display the dataset structure using info()

Find the number of rows and columns

Identify the data types of each column

ANSWER

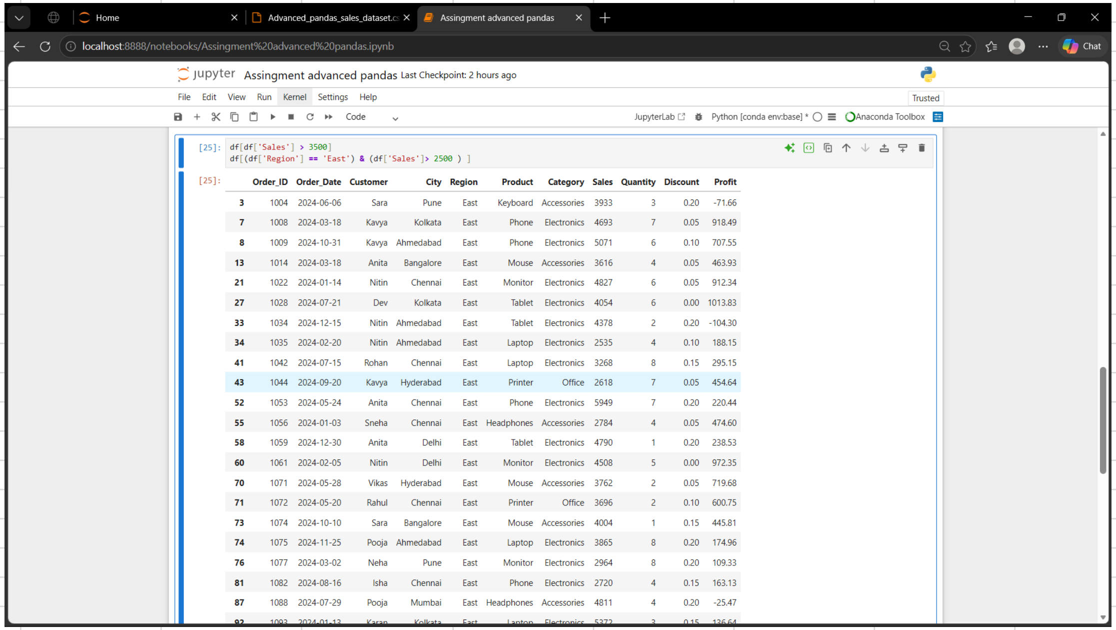


```
[6]: df.head(10),df.shape,df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 230 entries, 0 to 229
Data columns (total 11 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Order_ID    230 non-null    int64
 1   Order_Date  230 non-null    object
 2   Customer    230 non-null    object
 3   City        230 non-null    object
 4   Region      230 non-null    object
 5   Product     230 non-null    object
 6   Category    230 non-null    object
 7   Sales       230 non-null    int64
 8   Quantity    230 non-null    int64
 9   Discount    230 non-null    float64
 10  Profit      230 non-null    float64
dtypes: float64(2), int64(3), object(6)
memory usage: 19.9+ KB

[6]: (   Order_ID  Order_Date Customer       City Region   Product     Category  \
 0      1001  2024-10-01    Priya  Hyderabad   East     Mouse  Accessories
 1      1002  2024-05-03    Sneha    Kolkata   East   Monitor  Electronics
 2      1003  2024-01-26    Anita     Mumbai   West     Phone  Electronics
 3      1004  2024-06-06     Sara       Pune   East  Keyboard  Accessories
 4      1005  2024-10-15    Rahul  Bangalore  South   Monitor  Electronics
 5      1006  2024-07-04    Pooja  Bangalore   West   Printer       Office
 6      1007  2024-06-24      Dev    Chennai   East   Monitor  Electronics
 7      1008  2024-03-18    Kavya    Kolkata   East     Phone  Electronics
 8      1009  2024-10-31    Kavya  Ahmedabad   East     Phone  Electronics
 9      1010  2024-12-25    Pooja  Ahmedabad  North     Phone  Electronics

    Sales  Quantity  Discount  Profit
 0   1735         7      0.00  307.52
 1    807         2      0.15   49.89
 2   2328         1      0.20   66.47
 3   3933         3      0.20  -71.66
 4   3550         2      0.20  146.30
 5   3073         8      0.20  191.58
 6   1170         5      0.20   22.83
 7   4693         7      0.05  918.49
 8   5071         6      0.10  707.55
 9   2711         8      0.00  477.32  ,
 (230, 11),
```

QUESTION 2

Q2 Sales Filtering Analysis
Filter the dataset to display:
Orders where Sales > 3500
Orders where Region = "East" and Sales > 2500
Explain what insights you observe.

ANSWER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```
[25]: df[df['Sales'] > 3500]
      df[(df['Region'] == 'East') & (df['Sales']> 2500 ) ]
```

[25]:

	Order_ID	Order_Date	Customer	City	Region	Product	Category	Sales	Quantity	Discount	Profit
3	1004	2024-06-06	Sara	Pune	East	Keyboard	Accessories	3933	3	0.20	-71.66
7	1008	2024-03-18	Kavya	Kolkata	East	Phone	Electronics	4693	7	0.05	918.49
8	1009	2024-10-31	Kavya	Ahmedabad	East	Phone	Electronics	5071	6	0.10	707.55
13	1014	2024-03-18	Anita	Bangalore	East	Mouse	Accessories	3616	4	0.05	463.93
21	1022	2024-01-14	Nitin	Chennai	East	Monitor	Electronics	4827	6	0.05	912.34
27	1028	2024-07-21	Dev	Kolkata	East	Tablet	Electronics	4054	6	0.00	1013.83
33	1034	2024-12-15	Nitin	Ahmedabad	East	Tablet	Electronics	4378	2	0.20	-104.30
34	1035	2024-02-20	Nitin	Ahmedabad	East	Laptop	Electronics	2535	4	0.10	188.15
41	1042	2024-07-15	Rohan	Chennai	East	Laptop	Electronics	3268	8	0.15	295.15
43	1044	2024-09-20	Kavya	Hyderabad	East	Printer	Office	2618	7	0.05	454.64
52	1053	2024-05-24	Anita	Chennai	East	Phone	Electronics	5949	7	0.20	220.44
55	1056	2024-01-03	Sneha	Chennai	East	Headphones	Accessories	2784	4	0.05	474.60
58	1059	2024-12-30	Anita	Delhi	East	Tablet	Electronics	4790	1	0.20	238.53
60	1061	2024-02-05	Nitin	Delhi	East	Monitor	Electronics	4508	5	0.00	972.35
70	1071	2024-05-28	Vikas	Hyderabad	East	Mouse	Accessories	3762	2	0.05	719.68
71	1072	2024-05-20	Rahul	Chennai	East	Printer	Office	3696	2	0.10	600.75
73	1074	2024-10-10	Sara	Bangalore	East	Mouse	Accessories	4004	1	0.15	445.81
74	1075	2024-11-25	Pooja	Ahmedabad	East	Laptop	Electronics	3865	8	0.20	174.96
76	1077	2024-03-02	Neha	Pune	East	Monitor	Electronics	2964	8	0.20	109.33
81	1082	2024-08-16	Isha	Chennai	East	Phone	Electronics	2720	4	0.15	163.13
87	1088	2024-07-29	Pooja	Mumbai	East	Headphones	Accessories	4811	4	0.20	-25.47
92	1093	2024-01-13	Karan	Kolkata	East	Laptop	Electronics	5372	3	0.15	136.64

INSIGHTS	expensive products
	regions generating large revenue

QUESTION 3

Q3 Customer Purchase Analysis

Identify the top 5 customers based on total sales.

Hint: Use GroupBy with aggregation

ANSWER

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localhost:8888/notebooks/Assingment%20advanced%20pandas.ipynb

223	1224	2024-01-07	Dev	Bangalore	East	Tablet	Electronics	4115	4
225	1226	2024-06-14	Anita	Ahmedabad	East	Laptop	Electronics	5647	2
228	1229	2024-09-04	Priya	Pune	East	Tablet	Electronics	2635	8

[32]:
```python
df.groupby('Customer')['Sales'].sum().sort_values(ascending=False).head(5)
```

[32]:
```
Customer
Nitin    59820
Raj      58416
Rahul    52291
Dev      48500
Isha     46818
Name: Sales, dtype: int64
```

QUESTION 4

Q4 Regional Sales Performance
Calculate the total sales generated by each region.
Then identify:
Which region generated the highest revenue?

ANSWER

```
[76]: df.groupby('Region')['Sales'].sum()

[76]: Region
      East     232675
      North    158210
      South    183108
      West     170227
      Name: Sales, dtype: int64
```

QUESTION 5

Q5 Product Performance Analysis
Calculate the total quantity sold for each product.
Which product appears to be the most popular?

ANSWER

```
[10]: df.groupby('Product')['Quantity'].sum().sort_values(ascending=False)
```

```
[10]: Product
Monitor       156
Phone         151
Tablet        150
Laptop        140
Mouse         122
Headphones    115
Keyboard      110
Printer       108
Name: Quantity, dtype: int64
```

```
[11]: df.groupby('Product')['Quantity'].sum().sort_values(ascending=False).max()
```

```
[11]: 156
```

QUESTION 6

Q6 Multi-Level Grouping
Group the dataset by: Region and Category
Calculate the total sales for each combination.
What insights can you derive from this analysis?

ANSWER

```
West     Accessories     64811
         Electronics     84128
         Office          21288
Name: Sales, dtype: int64
```

```
[23]: df.groupby(['Region','Category']).agg(Total_Sales = ('Sales','sum'))
```

[23]:

Region	Category	Total_Sales
East	Accessories	57646
	Electronics	160907
	Office	14122
North	Accessories	62052
	Electronics	70023
	Office	26135
South	Accessories	80914
	Electronics	86690
	Office	15504
West	Accessories	64811
	Electronics	84128
	Office	21288

INSIGHTS

You can identify which category generates the highest revenue in each region.
Different regions may prefer different product categories.

STATEGIES FOR BUSINESS

Categories with consistently high sales across multiple regions indicate:
strong market demand
profitable products

QUESTION 7

Pivot Table Creation
Create a pivot table showing:
Rows → Region
Columns → Product
Values → Total Sales
Which region sells the most laptops?

ANSWER

West	Accessories	64811
	Electronics	84128
	Office	21288

```
[ ]: #Pivot Table Creation
     #Create a pivot table showing:
     #Rows → Region
     #Columns → Product
     #Values → Total Sales
     #Which region sells the most laptops?
```

```
[25]: pivot = df.pivot_table(values ='Sales', index ='Region', columns ='Product', aggfunc = 'sum')
      pivot
```

[25]:

Product	Headphones	Keyboard	Laptop	Monitor	Mouse	Phone	Printer	Tablet
Region								
East	23899	10845	55568	31299	22902	39667	14122	34373
North	15450	20538	4540	14922	26064	30535	26135	20026
South	24111	23480	23029	19167	33323	18218	15504	26276
West	17159	38499	10826	18178	9153	27850	21288	27274

INSIGHTS

The pivot table provides a clear comparison of product sales across regions.

- top-selling products
- strongest regions
- weak-performing markets

QUESTION 8

Profitability Analysis
Calculate the average profit for each product category.
Which category is most profitable?

```
[34]: df.groupby('Category')['Profit'].mean()
```

```
[34]: Category
      Accessories    410.603086
      Electronics    396.462114
      Office         471.938462
      Name: Profit, dtype: float64
```

ANSWER

QUESTION 9

Discount Impact Study
Analyze the relationship between discount and profit.
Tasks:
Calculate the average profit for each discount level
Identify whether higher discounts lead to lower profit

ANSWER

```
[38]: df.groupby('Discount')['Profit'].mean().sort_values(ascending=False)
```

```
[38]: Discount
      0.00    751.210588
      0.05    531.368039
      0.10    346.460513
      0.15    237.191400
      0.20     90.026923
      Name: Profit, dtype: float64
```

QUESTION 10

End-to-End Data Wrangling Task

Perform the following workflow:

Filter the dataset where Sales > 3000

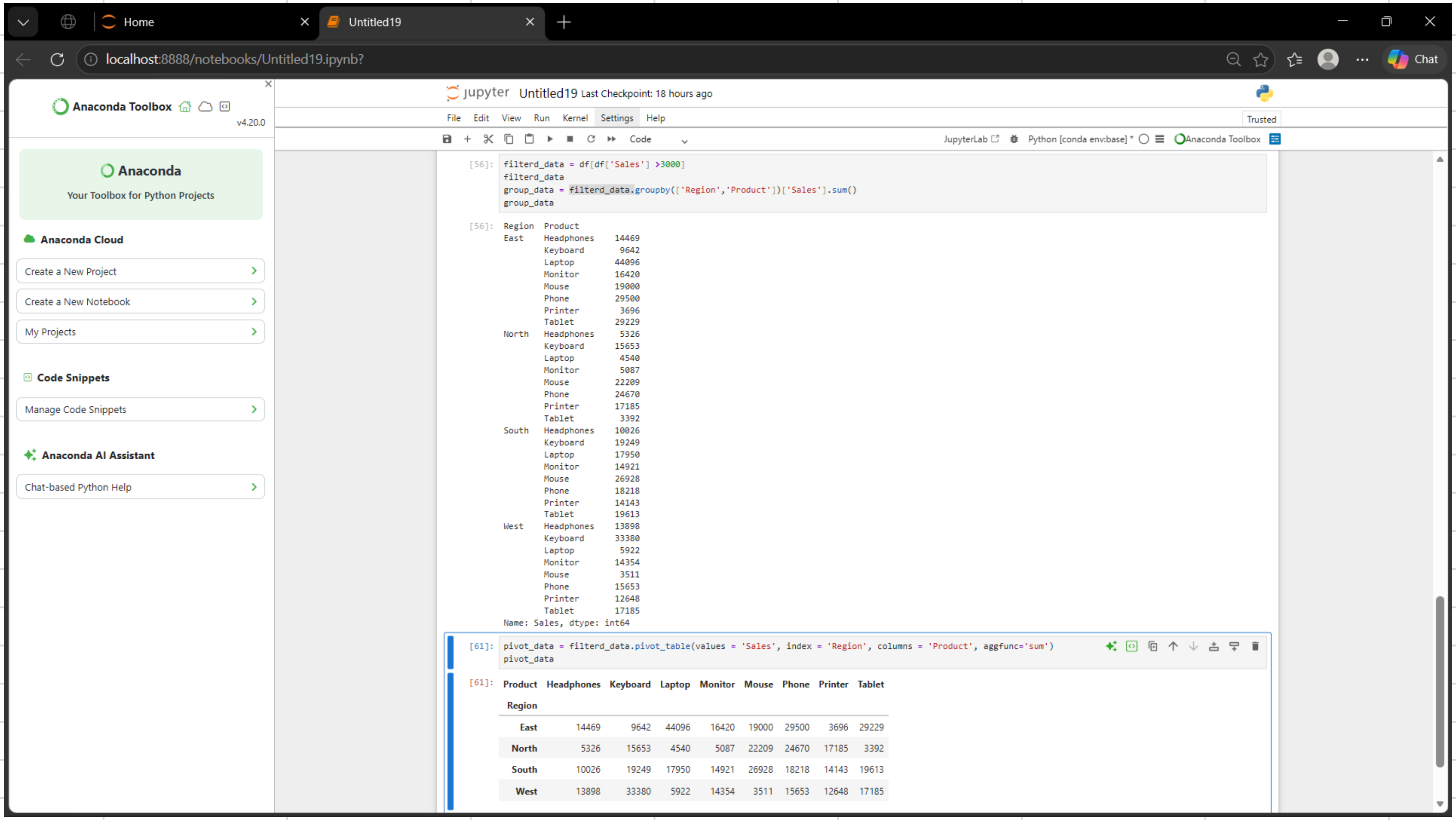
Group data by Region and Product

Calculate total sales

Reshape the result into a pivot table

Explain the business insights obtained from this analysis.

ANSWER



	The analysis shows which products generate the highest sales in each region.
INSIGHTS	Laptops may perform strongly in the West region.
	Different regions may have different product preferences.
	Target regional demand
	Since only orders above 3000 are analyzed
	The business can focus on premium customers